Champollion, Bumford, and Henderson on donkeys

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1. Donkey sentences and their ∀- and ∃-readings

(1) Every farmer who owns a donkey beats it
∀-*reading:* Every donkey-owning farmer beats every donkey he owns
∃-*reading:* Every donkey-owning farmer beats some donkey he owns

(2) No farmer who owns a donkey beats it
∀-*reading:* No donkey-owning farmer beats every donkey he owns
∃-*reading:* No donkey-owning farmer beats any donkey he owns

Note that for (1) the ∀-reading entails the ∃-reading, while for (2) the ∃-reading entails the ∀-reading.

CBH claim: both interpretations (and perhaps others) are available in principle for all donkey sentences (so long as the determiner is not an indefinite like 'a' or 'some'). But there are strong contextual effects on which interpretation we gravitate to for a given sentence.

(3) Everyone who had a quarter put it in the meter.
(4) No-one who had a quarter failed to put it in the meter.
(5) No-one who has an umbrella leaves it home on a day like this.

2. Ambiguity?

Ambiguous determiners theory: some determiners, including 'every', are ambiguous (whether lexically or by way of choices they force between different type-raisers). On one reading they generate the ∀ truth condition; on the other they generate the ∃ truth condition.

First objection: can the ambiguity theorist explain why context matters the way it does, and why the defaults are as strong as they are?

Second objection: it looks like (6) can have a reading intermediate in strength between the ∀-reading and the weaker ∃-reading, e.g. one that requires most farmers who own donkeys to beat most of their donkeys.

(6) Most farmers who own a donkey beat it

Third objection: mixed readings.

(7) Everyone who bought a book online and had a credit card used it to pay for it (Brasoveanu 2008).
Natural reading: Everyone who bought a book online and had a credit card paid for each book they bought online with some credit card they owned.

(8) Every man who has an umbrella takes it along on rainy days but leaves it home on sunny days (= ex. 53 in the paper)
Natural reading: Everyone who has an umbrella takes some umbrella they own along on every rainy day but leaves every umbrella they own at home on every sunny day.

A possible response to the last two objections: a 'rampant ambiguity' theory. There is a continuum of interpretations for each donkey sentence—every proposition entailed by the conjunction of the ∀- and ∃-readings and entailing their disjunction is itself a legitimate reading.

- Note that on some accounts of vagueness, vagueness is tantamount to ambiguity where the readings form a continuum.

3. The trivalence + QUD strategy

Idea: when a question (= partition of logical space) is salient in the right way ('under discussion'), it's OK to utter a trivalent sentence S so long as the true cell of the partition contains some worlds where S is true and no worlds where S is false, and no cell of the partition contains both worlds where S is true and worlds where S is false.

Thus, where ≈ is the same-cell relation generated by the QUD, the proposition *communicated* by an utterance of S will be

$$\{w\colon \exists w'(w' \approx w \text{ and } [\![S]\!](w) = \textbf{true})\}$$

Križ's application of this idea: explaining why 'The professors smiled' can be OK even when a few of them didn't smile.

4. A contrast between donkey sentences and plural definites

Križ makes a lot of the fact that we can't comfortably conjoin a plural definite sentence with something that entails that its strong reading isn't true.

(9) a. # Although the professors are smiling, one of them isn't.
b. # The professors are smiling, but one of them isn't.

Kris's explanation: the second conjunct makes the question whether one of the professors isn't smiling salient; but if it's part of the QUD, no world where the second conjunct is true is ≈ a world where the first conjunct is true.

The weak readings of donkey sentences, when available, don't seem so fragile.

(10) Everyone who had a quarter put it in the meter, and most had many quarters left over.

(11) Some people had many credit cards, some only had one, and some didn't have any. Everyone who had a credit card paid his bill with it.

5. Generating the required trivalent semantics for sentences

CDRT: there's a type 's' for indices/discourse referents (could be Nat), a type 'e' for objects, and a type 't' for truth values Assignment functions are type ⟨se⟩. Semantic values for sentences are type **t** = ⟨⟨se⟩⟨se⟩t⟩. Other expressions change to fit.

i[n]o means: i and o are assignment functions that agree on every index except n.

> $[\![\text{a man}_1\text{ came in}]\!] = \lambda io.i[1]o \wedge \mathsf{camein}(o(1))$
> $[\![\text{he}_1\text{ sat down}]\!] = \lambda io.i{=}o \wedge \mathsf{satdown}(i(1))$
> $[\![\text{farmer who owns a donkey}_1]\!]{=}\lambda xio.\mathsf{farmer}(x) \wedge i[1]o \wedge \mathsf{owns}\ x\ o(1) \wedge \mathsf{donkey}\ o(1))$
> $[\![\text{beats it}_1]\!] = \lambda xio.i{=}o \wedge \text{beats } x\ i(1)$

Determiners are born with type ⟨⟨et⟩⟨et⟩t⟩ and get lifted to ⟨⟨e**t**⟩⟨e**t**⟩**t**⟩ by a type shifter.

An implementation of the ambiguity theory in this framework: we have a choice between two type shifters E and U:

> $\underline{E}([\![\text{det}]\!]) = \lambda RNio.i{=}o \wedge [\![\text{det}]\!](\lambda x.\exists j.Rxij)(\lambda x.\exists j.Rxij \wedge \exists k.Nxjk)$
> $\underline{U}([\![\text{det}]\!]) = \lambda RNio.i{=}o \wedge [\![\text{det}]\!](\lambda x.\exists j.Rxij)(\lambda x.\forall j.Rxij \rightarrow \exists k.Nxjk)$

CBH's alternative: we just have one type-shifter D that generates trivalent type-**t** values.

> $\underline{D}([\![\text{det}]\!]) = \lambda RNio.i{=}o \wedge (E([\![\text{det}]\!])RNio * U([\![\text{det}]\!])RNio)$, where

*	**T**	#	**F**
T	T		#
#			
F	#		F

Upshot:

> $[\![\text{det farmer who owns a donkey}_1\text{ beats it}_1]\!]$ i o =
> **T** if i=o and ⟦det⟧(donkey-owner)(beater of every donkey one owns) and ⟦det⟧(donkey-owner)(beater of some donkey one owns)
> **F** if i≠o or ¬⟦det⟧(donkey-owner)(beater of every donkey one owns) and ¬⟦det⟧(donkey-owner)(beater of some donkey one owns)
> # otherwise

Note: the way CBH state the semantics assumes that the inputs to D⟦det⟧ are bivalent (i.e. map every i and o to either **true** or **false**. To treat sentences where donkey sentences occur under quantifiers, we'll have to relax this — the 'birth' semantic values of determiners will have to know what to do when their inputs map some objects to **neither**.

6. First overgeneration worry: too many communicative possibilities

The proposal allows for donkey sentences to be used to communicate all kinds of propositions strictly between the conjunction and the disjunction of the ∃- and ∀-readings, so long as the context can provide an appropriate QUD. For example (1) should be able to communicate 'every farmer who has a donkey beats most of his donkeys', or 'every male farmer who has a donkey beats all of his donkeys and every female farmer who has a donkey beats at least one of her donkeys', or many other things. Can it be used in these ways?

Even if we do everything we can to make these intermediate question salient it doesn't seem to work. For example: 'I heard that Peter said that every donkey-owner in this village beats most of his donkeys. That's interesting—I wonder if he's right? — Yes, every farmer who owns a donkey beats it'.

- CBH are aware there's an issue here and suggest a strategy for addressing it:

  > A further possibility is that the relationship between QUDs and the sentences that address them is constrained by question-answer congruence, that is, the notion that the answer and its focus alternatives must match the possible answers of the question (e.g. Roberts 2012). This might be the reason why (12a) ['Every student who took a course from Peter last year liked it'] does not lead hearers to accommodate QUDs such as *Did all of Peter's students like more than half of the classes they took with him?* and cannot be used to answer such questions in the affirmative.

- But I don't see how this can work and still keep the ability of the account to account for the intermediate readings we seem to get for sentences like 'Most farmers who own a donkey beat it' (which seems like it *can* mean something tantamount to 'most donkey-owning farmers beat most of their donkeys).

7. Second overgeneration worry: defaults

For lots of these sentences there seem to be pretty strong defaults—e.g. for the ∀-reading of (1) and (even stronger!) the ∃-reading of (2). CBH propose explaining this by saying that when we hear the sentences out of the blue we assume a 'fact-finding' QUD where it never happens that $w \approx w'$ where the sentence takes different truth values at w and w′. Such a QUD will generate the strongest possible communicated content, equal to the conjunction of the ∀- and ∃-readings.

This strategy does not explain what's going on with (12), where the default reading is the (weak) ∀-reading.

(12) Not every farmer who has a donkey beats it.
∀-reading: not every donkey-owning farmer beats every donkey he owns, i.e.

some donkey-owning farmer fails to beat some donkey he owns.
∃-reading: not every donkey-owning farmer beats some donkey he owns, i.e. some donkey-owning farmer fails to beat every donkey he owns.

This is not some idiosyncratic thing about the processing of 'not'. To the extent that there's a preference for the strong reading of a donkey sentence, sentences that embed that sentence seem to be understood as if the embedded occurrence had that strong reading, even when this leads to a weaker reading for the entire sentence.

(13) If every student who took a class from me liked it, I will get a bonus.
(14) If no student who took a class from me liked it, I will be fired.

This looks to me like a reason to prefer some kind of ambiguity view, where the 'readings' really are inputs to compositional semantics, to a CBH-style view where the multiplication of 'readings' is due to a process that comes along after the compositional semantics is done and is only sensitive to its output.

8. First undergeneration worry: super-weak readings

Does (14) (cf. example 8) have a "super-weak" reading, where it merely requires that every umbrella-owner takes some umbrella or other along on most rainy days?

(15) Everyone who has an umbrella takes it along most rainy days

I can't see any way of extending CBH's semantics that wouldn't predict that (14) is false in a world where some umbrella owners alternate among many umbrellas. (Note that we can't scope out 'most rainy days' since the intended reading is compatible with there being no rainy days when every umbrella owner has an umbrella with them.)

9. Second undergeneration worry: 'Nothing is F and G' versus 'No F thing is G'

I'm not seeing any difference between (16) and (17):

(16) No-one who has an umbrella left it home today
(17) No-one has an umbrella and left it home today.

Both seem to have the ∀-reading (which only requires that no umbrella-owners leave all their umbrellas at home).

Q: are (18)/(19) any different? (I'm not sure).

(18) No-one has an umbrella that they left at home today
(19) No-one left an umbrella at home today.

I don't see any plausible way to use CBH's machinery to generate for (17) (let alone (18) or (19)) the trivalent semantic profile they assign to (16).

- Their fragment doesn't include 'and', but one would expect something like

$$[\![\text{and}]\!] = \lambda PQio.\exists j(Pij \wedge_{\text{strong Kleene}} Qjo)$$

This will make ‘Fred met a girl and liked her’ equivalent to ‘Fred met and liked a girl’, and ‘Fred has an umbrella and leaves it at home’ equivalent to ‘Fred has an umbrella which he leaves at home’, etc.

- Could we instead postulate a semantics for ‘and’ modelled on CBH’s treatment of the quantifiers—i.e. a clause that makes ‘and’ equivalent to the $*$-combination of the above—call it $[\![\text{and}_\exists]\!]$—with $[\![\text{and}_\forall]\!] = \lambda PQio.\exists j(Pij \wedge \forall j(Pij \rightarrow Qjo))$.
- This will make ‘Fred met a girl and liked her’ is true if Fred met a girl and liked every girl he met’; false if there’s no girl Fred met and liked; otherwise neither true nor false. Seems dubious!